```c
#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
    int done;
};

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        p[i].done = 0;
    }

    int time = 0, completed = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    while(completed < n) {
        int min = -1;

        for(int i = 0; i < n; i++) {
            if(!p[i].done && p[i].at <= time) {
                if(min == -1 || p[i].bt < p[min].bt)
                    min = i;
            }
        }

        if(min == -1) {
            time++;
        } else {
            p[min].rt = time - p[min].at;
            time += p[min].bt;
```

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            p[min].ct = time;
            p[min].tat = p[min].ct - p[min].at;
            p[min].wt = p[min].tat - p[min].bt;
            p[min].done = 1;
            completed++;

            avg_tat += p[min].tat;
            avg_wt += p[min].wt;
            avg_rt += p[min].rt;
        }
    }

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
    for(int i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
               p[i].pid, p[i].at, p[i].bt,
               p[i].ct, p[i].tat, p[i].wt, p[i].rt);
    }

    printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Response Time = %.2f", avg_rt/n);

    return 0;
}
```

C:\Users\Admin\Desktop\1

```
Enter number of processes: 4

Process 1
Arrival Time: 3
Burst Time: 4

Process 2
Arrival Time: 2
Burst Time: 6

Process 3
Arrival Time: 1
Burst Time: 4

Process 4
Arrival Time: 3
Burst Time: 7

PID     AT      BT      CT      TAT     WT      RT
1       3       4       9       6       2       2
2       2       6       15      13      7       7
3       1       4       5       4       0       0
4       3       7       22      19      12      12

Average Turnaround Time = 10.50
Average Waiting Time = 5.25
Average Response Time = 5.25
Process returned 0 (0x0)   execution time : 28.697 s
Press any key to continue.
```

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SJF(NP)